

Data & AI Case Study

Mapped to two Wuerth internship roles

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This is my independent case study. I mapped the requirements of two Wuerth Data & AI internship postings to work I have already built and measured in my own portfolio (now 23 repositories, including the decision-chain integration capstone -- one real dataset through the whole distributor chain with 13 machine-checked reconciliation identities plus four additive ones -- an MCP agentic-integration server with contract-enforced result provenance and idempotent result caching, two warehouse-logistics flagships, a supply-network optimizer, an energy forecasting + dispatch study, a visual quality-inspection study with SPC monitoring, a fraud-operations study with gated retrain promotion, a predictive-maintenance policy study, and a live installable quantum-explainer PWA). The euro and accuracy figures throughout are on synthetic data (except retail-analytics-real and decision-chain, measured and labelled on real public data) and exist to prove the methods work and are measurable -- not to forecast Wuerth outcomes.

38

requirements mapped -> repo + artifact + measured proof

computed at build time from this repository's own guards: skill coverage 7/7 checks and consistency 9/9 checks passing; 23 portfolio repositories registered

Job #1 -- (Agentic) Automation with Low-code Platforms

Agentic AI workflows, low-code (n8n / Power Automate), connecting systems & APIs, document automation, ROI, rapid prototyping.

Job #2 -- Data & AI Analytics

BI / Power BI, KPI dashboards, forecasting & predictive analytics, Python & SQL, data modelling, turning data into decisions.

Who Wuerth is (public info)

Wuerth (public profile, approximate): a large family-owned German-HQ group in assembly/fastening and industrial MRO distribution -- ~400+ companies in 80+ countries, ~87,000 employees, ~EUR 20B+ revenue, a catalogue in the millions of articles, multi-channel (field sales, branches, e-commerce, e-procurement/EDI, ORSY inventory systems). A business that runs on assortment, pricing, availability, logistics, and high-volume transactional documents -- a natural fit for applied Data & AI.

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02

Opportunity map (summary)

Wuerth business area (public-info) -> Data/AI approach -> repo + measured result (synthetic unless noted)

Area (public-info)	Approach	Repo	Result (synthetic -- labelled if real)
Procurement / assortment / cross-sell	MILP vs greedy; Apriori/FP-growth; rule redundancy pruning	revops-optimizer, market-basket-analysis	MILP > greedy; 254 rules, top lift 2.41; 41% of rules redundant -- the 149 kept carry all the information
Pricing & margin	Elasticity + leakage; endogeneity fix; per-move robustness gate; ablation	revops-optimizer, sales-kpi-analytics, ml-models-lab	EUR2.6M leakage lever; bias +1.52 -> +0.03; price-move gate 17/29 ACCEPT -- the biggest move (45% of the pricing uplift) is HELD, stated as a screen; ablation: endogeneity-control knockout +1.446 RMSE -- the load-bearing piece; scenario compare: EUR159,966.19 -> EUR139,487.29 attributed to named drivers with a EUR0.00 residual -- pricing gains EUR7,382.88 while the assortment gap loses EUR23,895.20, and its two ORDERED drivers are disclosed
Inventory / replenishment	Newsvendor + forecast + ROP/EOQ policy + supplier reliability	distributor-intel-platform, ml-models-lab	MASE 0.38; MASE 0.987 / RMSSE 0.948 (beats naive + Holt-Winters); policy: EUR127,421 working capital (cycle+safety exactly), 5.5x turns, 99.9% fill; measured lead times cost +14.4% safety stock -- wobble is 3/4 of the bill
Sales KPIs & analytics	KPIs + rolling-origin CV + pacing interval + rep decomposition	sales-kpi-analytics	MASE < 1; exec PDF; pacing EUR10.91M = 95.7% of plan, 80% interval -- closed-year back-check landed above the band, reported; rep decomposition (indirect standardization): 92% of the league table is territory, 8% execution; concentration: HHI 68 company-wide but 2,064 inside one territory (the largest account is 35.76% of one rep's book) -- removal arithmetic on measured revenue, no churn modelled anywhere
Logistics / routing	OR-Tools CP-SAT VRP + robustness + Fleet Size and Mix VRP	route-optimizer	4.6% / 31% savings; 5% headroom: failing scenarios 96% -> 44%, expected day -4.6%; fleet mix: -17.5% vs the status quo with the service price shown (longest route +25.2%); driver shifts: the statutory limits are SLACK on this data (worst duty 267 of 600 min, 43 min of continuous-drive slack, zero breaks needed) -- capping at 210 min costs +17.5% km and an eleventh van, 180 min is infeasible at any fleet size; informed by EU 561/2006, not a compliance tool

Opportunity map (summary) (continued)

Area (public-info)	Approach	Repo	Result (synthetic -- labelled if real)
E-procurement automation	Agentic + n8n low-code + reliability benchmark + content-verification layer + dry-run flow cost estimator	agentic-automation-lab, agent-flow-studio	~EUR625k/yr modelled; 1,350 seeded trials: content faults fail silently, stated (assumed rates); verification layer on the same fault schedule: 88.7% -> 98.8% delivered-correct, silent-wrong 11,523/yr -> 0 (price stated; 100 tests); flow estimator: engine-verified branch scenarios, 'not a bill' (declared rates); HITL checkpoints priced per placement: risk-gated review takes silent-wrong 11,523/yr -> 0 for EUR88,948/yr = EUR7.65 per prevented error, per-step approval costs EUR349,708/yr and buys nothing extra, and a skipped validation is MISSED at pre-commit -- the reviewer is modelled generously, so catch rates are upper bounds
Document processing	Agentic extract + 7-rule validation gate + cost model	doc-extract-agent	~EUR145k/yr modelled; combined-gate precision 70.0% -> 87.5%; cost model: auto-post pays only above 98.4% precision -- the gate alone would lose money (modeled), the pre-fill carries the value; source spans: every read value underlined where it was read (25 of 26 located, 1 derived marked 'no span'), verified across all 30 committed documents -- a character range, never a pixel region
Customer retention	Decline + churn classifiers + ablation	revops-optimizer, ml-models-lab	ROC-AUC 0.99; churn ECE 0.197 -> 0.021; bootstrap skill CI +0.457 [+0.381, +0.528]; ablation: freq_slope knockout -0.247 PR-AUC; null knockouts not counted as wins
Warehouse / WMS (new)	WMS twin + factory/plant sim (Story Mode, 933-element plant, definable objects, multi-way line sim + fluids solver); slotting + DES + packing + pick-path routing + order batching	logistics-flow-studio (WarehouseTwin v3.38), logistics-digital-twin (engine)	-48.6% pick travel; ABC ~21% > random; ISO 22400 KPIs; per-order routing (20 ops, 8 archetypes); 178/178 self-test + 70 harnesses + 87 Python tests; line sim 112.5 parts/hr (~88.1% eff.); fluids solver (modelled); engine slotting -44.2% (golden-zone 25% -> 100%); fill 2.0% -> 30.2%; routing: return +3.0% vs exact optimum, optimized layout ~46% shorter; batching: savings -71.3%, routing flips to largest-gap; v3.21-v3.24 plant floor -- concrete + safety-yellow paint, living workers, pallet -> carton -> tote -> parcel, hauling forklifts, anti-strobe congestion bands, dock trailers (RENDERING, not model: no number, no export changes); floor plan: ABC-classed bays, a real pick tour 69.2 m vs its 64.4 m exact optimum (+7.5%)
Supply-network design (new)	Facility MILP + flows + safety stock + service frontier + growth plan	supply-network-opt	-21.2% cost vs greedy; stock -65.7% / -80.1%; frontier \$2,353 -> \$14,750 per service point (6.3x); growth: +8.8% headroom, 4th DC first pays at 1.30x (planning estimates); phased build: staged \$2,750,462 NPV (DC0 yr 2, DC3 yr 6) vs a \$2,598,732 free-redesign LOWER BOUND, building ahead +24.4%, no expansion fails in year 2 -- growth and discount rates illustrative, fixed cost as recurring opex not capex, no construction lead time

Opportunity map (summary) (continued)

Area (public-info)	Approach	Repo	Result (synthetic -- labelled if real)
Energy management / facilities (new)	Load forecast (rolling CV) + peak-shaving LP + causal dispatch backtest + degree-day decomposition	energy-demand-forecast	MASE 0.497, 14/14 folds (Holt-Winters loses to naive, reported); peak -20.9%; ~EUR11,100/yr at ASSUMED tariff; battery does not pay back on these assumptions; backtest captures 72.7% of the LP bound (robust variant loses, reported); decomposition recovers designed balance points exactly -- weather 4.5% of energy, 18% of the July peak (modelled attribution); battery sizing: at an ASSUMED EUR375/kWh the 15-year break-even is EUR25/kWh-yr and only the 100 kWh system clears it (11.3-yr payback vs 18.6 at 400 kWh) -- and the linear per-kWh price is flagged as a caveat that cuts against the repo's own headline
Quality inspection (new)	Clean-only anomaly detection; pre-registered rule; SPC p-chart + WE rules; measured OCAP	quality-anomaly-vision	AE 0.779 vs PCA 0.772 ROC-AUC -- inside 0.02 margin, PCA recommended; TPR 0.407 vs 0.393 @ 5% FPR; calibration correction 0 -> 0.70% measured; OCAP: label-free re-centering is a trap (green chart, near-blind screen); refit on verified-clean frames recovers ~99% of the drift cost; severity grading: the bill rises 368 -> 509 EUR/1,000 parts (+39%) but THE DECISION DOES NOT MOVE (a critical escape needs 259 EUR, 7.4x, to shift it), and the expensive grade is the one the screen sees worst (critical AUC 0.706)
Real data (completed)	Cleaning + RFM + returns + lifecycle + leakage-safe CV	retail-analytics-real	real data: seasonal-naive wins CV, MASE 1.094; returns 3.65% of gross, 95.0% of value matched, median 10-day lag; Champions 25% -> 69.0%; lifecycle: resurrections outnumber repeats (439 vs 390/month); price ladders: 88.7% of SKUs sold at more than one price, realization 98.6% ('not an opportunity'), slopes -1.71 within-week / -1.67 market-adjusted with 74.1% beating a permutation null -- and NO elasticity claimed, nothing causal
Chain integration & reconciliation (new)	Provenance-tagged pipeline + identity ledger; MCP agentic layer w/ machine-readable result provenance + idempotent caching	decision-chain, chain-mcp	real data + labelled layers: 13/13 identities PASS + additive (n), (o), (p), (q); per-order spread to the cent over 4,151 orders (top decile 59.0%, Gini 0.665); fleet knob: only the transport line moves, vans unpriced (model property, not fleet advice); cross-repo revenue GBP 19,643,861.62 to the penny; naive wins lumpy (MASE 1.782); contract-enforced provenance + cache-status, 193 tests; plan diff (synthetic): EUR136,972.20 -> EUR115,728.86 with every euro attributed to a named cause over 11 change identities and a -EUR0.02 rounding line against a EUR4.03 bound -- real, never fitted to close the gap
Fraud & transaction-risk ops (new)	Cost-based alerting; gated retrain promotion; selective-labels feedback sim	fraud-detection-ops	PR-AUC 0.270 vs oracle 0.367; swap-set gates -> PROMOTE at \$8,632 vs \$8,841 -- retrain finds no new fraud, wins by shedding load (stated); feedback sim: ranking survives censoring, probabilities + alerts collapse; reason codes: EXACT Shapley on the linear champion, checked against brute-force enumeration over all 2^m coalitions -- and the most common reason is not the most predictive one (merchant_category principal on 40.5% but 11.4% fraud vs transaction_amount 19.1% / 20.7%); logits against a stated reference, not causal, not legal advice

Opportunity map (summary) (continued)

Area (public-info)	Approach	Repo	Result (synthetic -- labelled if real)
Maintenance & asset reliability (new)	Censored Weibull + age-replacement vs condition-based + CBM re-tuning	predictive-maintenance	beta 4.81; $T^* = 44.4$ d at 7.16/machine-day -- the calendar rule beats the repo's own detector at the default threshold, reported; re-tuned economically the ranking flips to CBM (4.02) -- bounded, in-sample; spares from the same fit: run-to-failure/CBM 0.0148 parts/machine-day (base stock 126) vs age replacement 0.0229 (188) -- the calendar rule draws +54% MORE PARTS while cutting the cost rate 51.7%; a modelled provisioning exercise, not an order

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03

Job #1 -- (Agentic) Automation with Low-code

Every posting requirement -> repo + artifact + measured proof (all synthetic data).

Requirement	Repo / artifact	Proof (synthetic unless noted)
Build AI-agent workflows -- and measure how they fail	agentic-automation-lab (+ eval/reliability.py benchmark + eval/verification.py content-verification layer)	agentic tool-use loops; reliability benchmark (1,350 seeded trials): content faults fail silently past the guards and retries can't zero them (assumed fault rates, stated); content-verification layer replays the same fault schedule: delivered-correct 88.7% -> 98.8%, silent-wrong 11,523/yr -> 0 on that schedule (escalations 260 -> 1,300/yr stated; prose blind spot unit-tested); and the review itself is now priced where it sits -- risk-gated approval takes silent-wrong to 0 for EUR88,948/yr (EUR7.65 per prevented error) while per-step approval costs EUR349,708/yr and buys nothing extra; 100 tests
Agentic integration via an open standard (MCP)	chain-mcp (official mcp Python SDK; Claude Desktop / Code configs)	6 real engines as AI-callable tools; typed schemas; never-crash errors; contract-enforced machine-readable result provenance (data label + engine commit + determinism flag); idempotency + result caching -- the key binds args + the exact engine checkout, and every success says computed-or-replayed (provenance.cache, contract-required); 193 tests incl. live JSON-RPC handshake
Low-code (n8n / Power Automate)	agentic-automation-lab: n8n/rfq_intake_agent.json	RFQ-intake agent; ~EUR625k/yr
Visual / flow-based building	agent-flow-studio (+ estimator.js dry-run estimator)	flow builder; ~EUR47k/yr; dry-run cost/latency estimator: declared rate card, branch scenarios enumerated and proven against real engine runs branch-for-branch -- 'emphatically not a bill' (78 node tests)
Connecting systems / APIs	agentic-automation-lab connector layer	agents call external APIs
Process automation (back-office)	doc-extract-agent	RFQ/invoice -> structured
Document intake / understanding	doc-extract-agent + 7-rule validation layer + cost model (eval/run_cost)	extract + validate; ~EUR145k/yr; combined-gate precision 70.0% -> 87.5%; cost model: auto-post pays only above 98.4% precision -- the gate alone would lose money (modeled), the pre-fill carries the value; source spans: every read value underlined where it was read (25 of 26 located, 1 derived marked 'no span'; a non-matching span is reported as NOT LOCATED), verified across all 30 committed documents
Rapid prototyping	agent-flow-studio, agentic-automation-lab, logistics-flow-studio	runnable prototypes incl. a full installable offline WMS twin + factory/plant simulator (PWA, v3.38; 70 harnesses + 87 Python tests + 178/178 self-test), whose floor now reads like a working shift -- rendering, not model

Job #1 -- (Agentic) Automation with Low-code (continued)

Requirement	Repo / artifact	Proof (synthetic unless noted)
Prototyping + education (hype-free)	quantum-explainer -- LIVE: dimitres-kisimov.github.io/ quantum-explainer; bio-efficient-ai (learned BioHash vs random, public MNIST benchmark)	hand-written simulator ~300 lines, zero deps; lessons incl. superdense coding (no-FTL demo -- the encoded pair is locally invisible; Holevo's bound respected, not beaten); 201 physics assertions + 53 self-test + 93 structural checks pass; BioHash: mid-range wins from a 10x smaller circuit -- negatives kept (both tips lost, no memory-frontier point, corpus-dependent)
Human-in-the-loop control of agentic actions (new row)	agentic-automation-lab: src/agentic_lab/checkpoints.py + eval/checkpoints.py	where the human sits, priced. A gate hands out a resumable, digest-stamped envelope, so an approved gate costs 0 extra tokens (restarting would cost 7.90x more) and an edited envelope is refused. Five placements on the same fault schedule: none -> 11,523 silent-wrong/yr; risk-gated -> 0 for EUR88,948/yr = EUR7.65 per prevented error; pre-delivery EUR188,438/yr; every step EUR349,708/yr and buys nothing extra. Placement changes COVERAGE too -- a skipped validation is missed at pre-commit and every-step. The reviewer is modelled generously, so catch rates are upper bounds and the zero false-alarm rate says nothing about people; EUR labour and USD tokens reported side by side, never summed; 100 tests
Show automation ROI	automation-roi-explorer	EUR383k/yr net modelled
Python engineering	all automation repos	Python throughout
Stakeholder communication	automation-roi-explorer	business-readable ROI views

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04

Job #2 -- Data & AI Analytics

Every posting requirement -> repo + artifact + measured proof (synthetic data unless labelled real data).

Requirement	Repo / artifact	Proof (synthetic unless noted)
One dataset through the whole chain (real data + labelled layers)	decision-chain (run artifact + offline dashboard; FAIL path per identity)	13/13 reconciliation identities PASS + additive (n), (o), (p), (q); cross-repo revenue GBP 19,643,861.62 to the penny; ledger to the cent; forecast-error elasticity == holding share exactly (0.0580); per-order cost-to-serve to the cent over 4,151 real orders -- top decile 59.0%, Gini 0.665 (model shape, not profitability); fleet knob swept with every day's CVRP re-solved: only transport moves, vans unpriced (not fleet advice); naive wins lumpy (MASE 1.782); slotting optimum -1.6% vs ABC; CVRP -0.2% vs Clarke-Wright; cost rates labelled INVENTED
BI / Power BI	revops-optimizer: powerbi/DAX_measures.md + kpi_uplift_risk	DAX pack + 3-page report + Monte-Carlo P10/P50/P90 risk table w/ DAX
KPI dashboards / metrics + fair performance measurement	sales-kpi-analytics + pacing bullet chart + repperf.py	KPI layer + exec PDF; pacing EUR10.91M = 95.7% of plan, 80% interval; back-check: realised EUR11.28M landed above the band -- honest miss; rep decomposition (indirect standardization): 92% of the league-table dispersion is territory, only 8% execution; concentration: Gini 0.5476, HHI 68 company-wide (~148 effective accounts of 383) but 2,064 inside one territory, where the largest account is 35.76% of that rep's book -- removal arithmetic on measured revenue, no churn probability modelled
Uncertainty / risk quantification	revops-optimizer simulate.py + robustness.py; ml-models-lab BENCHMARK_CI.md	P10-P90 EUR145,091..170,723/yr, ~43% chance of clearing the headline; tornado ranks drivers; price-move gate: 17/29 ACCEPT, the biggest move HELD (67% carry) -- a screen, not a guarantee; bootstrap skill CIs: churn +0.457 [+0.381, +0.528]; scenario compare: EUR159,966.19 -> EUR139,487.29 with every euro attributed to a named driver and a EUR0.00 residual (pricing gains EUR7,382.88 while the assortment gap loses EUR23,895.20) -- two ORDERED drivers disclosed, the gap shipped whole; 91 tests
Inventory policy / working capital	distributor-intelligence-platform dip/inventory.py + dip/reliability.py; supply-network-opt frontier	ROP/EOQ at ABC-XYZ targets: EUR127,421 working capital (cycle+safety exactly), 5.5x turns, 99.9% fill; supplier reliability: measured lead times cost +14.4% safety stock -- wobble is 3/4 of the bill; frontier 6.3x convexity
Predictive analytics / forecasting	sales-kpi-analytics; ml-models-lab global forecaster	MASE < 1; MASE 0.987 / RMSSE 0.948 beats naive + Holt-Winters
Forecasting rigour + reconciliation guard	distributor-intelligence-platform (MRO command center)	MASE 0.38 (9 rolling folds); /reconcile guard renders a green 'no silent drift' verdict over 16/16 cross-engine identities with all 21 headline numbers present; new plan-diff station: EUR136,972.20 -> EUR115,728.86 with every euro attributed to a named cause over 11 change identities -- price moves -EUR29,931.98, drops -EUR150,919.35 against greedy re-picks +EUR150,224.01, routing +EUR9,384.00 and a -EUR0.02 rounding line against a EUR4.03 bound, real and never fitted to close the gap

Job #2 -- Data & AI Analytics (continued)

Requirement	Repo / artifact	Proof (synthetic unless noted)
Energy forecasting + optimization	energy-demand-forecast (forecasters + LP + dispatch backtest + degree-day decomposition; 72 tests)	MASE 0.497 / MAPE 4.8%, 14/14 folds (Holt-Winters loses to naive, reported); peak 368.2 -> 291.1 kW (-20.9%); ~EUR11,100/yr at ASSUMED EUR12/kW-month; timer baseline EUR0; battery does not pay back on these assumptions; causal backtest captures 72.7% of the LP bound (robust variant loses); decomposition recovers designed balance points exactly -- weather 4.5% of energy, 18% of the July peak (modelled attribution, not a sub-meter); battery sizing on the same causal controller: break-even EUR25/kWh-yr at an ASSUMED EUR375/kWh, and only the 100 kWh system clears it (EUR33.3/kWh-yr, 11.3-yr payback) while 400 kWh buys 2.4x the saving for 4x the battery -- the linear per-kWh price is flagged as cutting against the repo's own headline
Python for analytics	sales-kpi-analytics, revops-optimizer	end-to-end pipelines
SQL / data modelling	sales-kpi-analytics SQL set	spend analysis queries
Excel-level tabular analysis	sales-kpi-analytics	spend breakdowns
Optimization / prescriptive	revops-optimizer; supply-network-opt; logistics-digital-twin	~EUR160k/yr; -21.2% cost vs greedy; fill 2.0% -> 30.2% (CP-SAT proof)
Elasticity / statistical modelling	revops-optimizer; ml-models-lab (+ ablation -> docs/ABLATION.md, drift-gated)	ROC-AUC 0.99; elasticity bias +1.52 -> +0.03; churn ECE 0.197 -> 0.021, bootstrap skill CI +0.457 [+0.381, +0.528]; ablation: freq_slope knockout -0.247 PR-AUC, endogeneity-control knockout +1.446 RMSE; the calibration knockout's bit-identical PR-AUC is the kept honest exception; null knockouts not counted as wins (numpy only, CI-reproducible)
Data into business decisions	sales-kpi-analytics; market-basket-analysis	EUR2.6M leakage lever; cross-sell recs, top lift 2.41
Market-basket / cross-sell	market-basket-analysis + affinity.py + redundancy.py	Apriori + FP-growth from scratch; 224 itemsets, 254 rules; affinity network Q=0.58, 3 communities, 4 bridges; 41% of rules redundant -- 149 carry all the information; 68 tests
Classification / anomaly detection	ml-models-lab; fraud-detection-ops	SKU macro-F1 0.963; AE PR-AUC 0.963 vs PCA 0.951 (PCA default); fraud PR-AUC 0.270 vs oracle ceiling 0.367; 55 tests
Model lifecycle / MLOps (gated retrain)	fraud-detection-ops challenger.py + feedback.py	swap-set + five pre-declared gates -> PROMOTE at \$8,632 vs \$8,841; retrain finds no new fraud, wins by shedding load (\$8/review assumption, stated); selective-labels feedback sim: ranking survives censoring (PR-AUC 0.268-0.273) but probabilities die -- alerts starve 651 -> 197 while the arm's own dashboard reads 100% recall by construction; 78 tests
Reliability / maintenance (survival)	predictive-maintenance pdm/policy.py + pdm/cbm_tuning.py	censored Weibull beta 4.81; age-replacement T* 44.4 d at 7.16/machine-day beats the repo's own detector (8.28) at the default threshold -- reported; re-tuned economically the ranking flips to CBM (4.02, break-even inspection cost 553.3) -- bounded, in-sample; spares from the same fit: run-to-failure/CBM 0.0148 parts/machine-day (base stock 126) vs age replacement 0.0229 (188) -- the calendar rule draws +54% MORE PARTS while cutting the cost rate 51.7%; Poisson checked, not assumed (cycle CV 0.24 / 0.07); a modelled provisioning exercise, NOT an order -- no lead time, no batching, counts units not money; 80 tests

Job #2 -- Data & AI Analytics (continued)

Requirement	Repo / artifact	Proof (synthetic unless noted)
Visual quality inspection (vision)	quality-anomaly-vision (3 detectors, pre-registered rule; SPC + OCAP; 59 tests)	AE ROC-AUC 0.779 vs PCA 0.772 -- inside 0.02 margin, PCA recommended; TPR 0.407 vs 0.393 @ 5% FPR; texture-breaks hard for all (best 0.609); SPC p-chart + WE rules; calibration correction 0 -> 0.70% measured; measured OCAP: label-free threshold re-centering is a trap (bit-identical ROC-AUC, 1.1/15 defects at +0.10 drift); refit on 200 verified-clean frames recovers ~99% of the drift cost; severity grading: the bill rises 368 -> 509 EUR/1,000 parts (+39%, 63% of it from 2.3 critical escapes) but the operating point does not move -- a critical escape needs 259 EUR (7.4x) before it does -- and critical AUC 0.706 vs 0.819 major: the expensive grade is the one the screen sees worst
Explainability / decision transparency (new row)	fraud-detection-ops: fdo/reasons.py (python -m fdo -- reasons)	reason codes in the shape regulated lending uses -- computed from the model already trained: no surrogate, no sampling, no randomness. The champion is linear in its standardized features, so against a stated reference profile ($z(r) = -0.509$) the per-feature terms ARE the Shapley values, asserted against brute-force enumeration over all 2^m coalitions rather than cited; the explained set is the same 608 alerts the shipped threshold fires. The finding is negative: the most common reason is not the most predictive one -- merchant_category is principal on 40.5% of alerts but confirms fraud only 11.4% of the time, while transaction_amount is principal on 19.1% and confirms 20.7%. Contributions are logits against a stated reference: not probabilities, not dollars, not causal -- and the four-reason format is borrowed from adverse-action practice as a discipline, not legal advice; 78 tests
Warehouse / intralogistics + factory / process industry	logistics-flow-studio (WarehouseTwin v3.38); logistics-digital-twin (engine)	WMS twin + factory/plant sim: keyword-generated layout, WMS flow w/ ISO 22400 KPIs, live material flow + KPI dashboard, Story Mode tour, a 933-element signature plant (32 object types), a user-definable object library, multi-way line sim (QA-split 112.5 parts/hr, ~88.1% eff.) + fluids solver (modelled, not measured), BYO CSV import + floor-plan underlay; -48.6% pick travel; ABC ~21% > random; per-order routing (20 ops, 8 archetypes); 178/178 self-test + 70 harnesses + 87 Python tests; engine slotting -44.2% (golden-zone 25% -> 100%), fill 2.0% -> 30.2%; DES -76.1% / -66.5%; pick-path routing: return +3.0% vs exact optimum, optimized layout ~46% shorter; batching: savings -71.3%, routing flips to largest-gap (+1.2%); v3.21-v3.24 plant floor -- industrial materials, living workers with per-station task poses, physical goods (pallet-load -> carton -> tote -> parcel) on belts and forks, hauling forklifts, frame-rate-invariant anti-strobe congestion bands, dock trailers and an andon lamp: RENDERING, not model -- no number, no export changes; and the engine's result drawn on the floor it happens on (a renderer, not a new model): ABC-classed bays, a real pick tour 69.2 m against its 64.4 m exact optimum (+7.5%), batching plate 254.0 -> 66.8 m (-73.7%)

Job #2 -- Data & AI Analytics (continued)

Requirement	Repo / artifact	Proof (synthetic unless noted)
Logistics / ops analytics	route-optimizer (74 tests, fleetmix.py + shifts.py); supply-network-opt (71 tests, growth.py + phasing.py)	4.6% / 31% savings; robustness: 5% headroom cuts failures 96% -> 44%, expected day -4.6%; fleet mix (FSM): 5 large vans -17.5% vs 10 mediums with the service price shown (longest route +25.2%); a genuine mix wins the small instance (-8.5%; illustrative catalogue); safety stock -65.7% / -80.1%; frontier 6.3x; growth plan: +8.8% headroom, 4th DC first pays at 1.30x (planning estimates); driver shifts: the statutory limits are SLACK on this data (worst duty 267 of 600 min, 43 min of continuous-drive slack, zero breaks needed) -- capping at 240 min costs +5.3% km, 210 min +17.5% and an eleventh van, 180 min is infeasible at any fleet size (a proof, not a search that gave up); informed by EU 561/2006, not a compliance tool. Phased build: staged \$2,750,462 NPV (DC0 yr 2, DC3 yr 6) vs a \$2,598,732 free-redesign lower bound, building ahead +24.4%, never closing +5.8%, no expansion failing in year 2 -- illustrative growth and discount rates, recurring opex not capex, no construction lead time
Real, messy data (completed)	retail-analytics-real (UCI Online Retail II, 1,067,371 real rows, CC BY 4.0)	real data: 94.0% rows kept; GBP 19.6M revenue; returns 3.65% of gross, 95.0% of value matched, median 10-day lag; Champions 25% -> 69.0%; lifecycle: resurrections outnumber repeats (439 vs 390/month); seasonal-naive wins CV, MASE 1.094; price ladders: 88.7% of SKUs sold at more than one price, realization 98.6% ('not an opportunity'), slopes -1.71 within-week / -1.67 market-adjusted with 74.1% beating a permutation null -- and NO elasticity claimed, nothing causal: an assortment-level statement, not a per-SKU price recommendation
Present to decision-makers	revops-optimizer exec deck	exec narrative deck